



Question 1) Neutrinos are fundamental particles that interact only via the weak force and gravity, making them extremely difficult to detect. The weak force is responsible for beta decays of neutrons and other unstable nuclei and it produces neutrinos in very specific combinations called **flavor states**: electron neutrino $|\nu_e\rangle$, muon neutrino $|\nu_\mu\rangle$, and tau neutrino flavor state $|\nu_\tau\rangle$. The key observation for this problem is that these “flavor” states are not eigenstates of the free particle Hamiltonian and therefore their time evolution is not a single trivial phase. This mismatch between the states produced in interactions and the true energy eigenstates gives rise to the phenomenon of **neutrino oscillations**, fully described by the time evolution formalism in quantum mechanics. In this problem, you will derive the probability for neutrinos to oscillate from one flavor to another as they propagate through space and look at some real and recent data I will work with natural units in all that follows ($\hbar = c = 1$).

a) Write the neutrino flavor state as a superposition of energy eigenstates:

$$|\nu_\alpha\rangle = \sum_{i=1}^3 U_{\alpha i}^* |\nu_i\rangle \quad (1)$$

with $U_{\alpha i} = \langle \nu_\alpha | \nu_i \rangle$ being the elements of the unitary matrix that relates/transforms the flavor eigenstates to the energy eigenstates (recall S&N Sec 1.5.1). We often refer to the energy eigenstates as the “mass eigenstates” because they have a well-defined mass (and therefore well-defined energy for a given momentum). Flavor states represent the specific superposition of mass eigenstates that are produced when neutrinos are created or detected via weak interactions alongside their corresponding charged leptons (electrons, muons, taus). The matrix U is called the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix and it is unitary, $U^\dagger U = U U^\dagger = \mathbb{1}$. Prove that

$$\langle \nu_\alpha | \nu_\beta \rangle = \delta_{\alpha\beta} \quad (2)$$

using the unitarity of the PMNS matrix and the fact that the mass eigenstates are orthonormal $\langle \nu_i | \nu_j \rangle = \delta_{ij}$.

b) Neutrinos are extremely light particles (one of them may be massless, we don’t yet know), so they are always ultra-relativistic, i.e., their momentum is much larger than their mass, $p \equiv |\vec{p}| \gg m_i$. In special relativity, their energy is given by

$$E_i = \sqrt{p^2 + m_i^2} = p \sqrt{1 + \frac{m_i^2}{p^2}} \approx p + \frac{m_i^2}{2p}. \quad (3)$$

We can take some take inspiration from the energy-momentum relation and build a Hamiltonian for a single mass eigenstate neutrino such that

$$\hat{H} |\nu_i\rangle = E_i |\nu_i\rangle = \left(p + \frac{m_i^2}{2p} \right) |\nu_i\rangle. \quad (4)$$

Note we are assuming all neutrinos mass states share the same momentum p . Find the time evolved state $|\nu_\alpha(t)\rangle$ for a neutrino initially in the flavor state $|\nu_\alpha\rangle$, given the Hamiltonian $\hat{H}_i$ for the mass eigenstates.

c) Now for a two-flavor neutrino system, for example $\{\nu_e, \nu_\mu\}$ flavor states and $\{\nu_1, \nu_2\}$ mass eigenstates, calculate the probability of a neutrino initially in the flavor state $|\nu_\alpha\rangle$ being detected in the flavor state $|\nu_\beta\rangle$ at time t , i.e., calculate both

$$P_{\mu e}(t) = |\langle \nu_e | \nu_\mu(t) \rangle|^2 \quad \text{and} \quad P_{ee}(t) = |\langle \nu_e | \nu_e(t) \rangle|^2 \quad (5)$$

Express your answer in terms of $|U_{\alpha i}|^2$, the mass-squared differences $\Delta m_{ij}^2 \equiv m_i^2 - m_j^2$, and time t . Express your answer in terms of sin functions.

(Hint: if you want to simplify your answer, try using the unitarity of U through relations like $|U_{e1}|^2 + |U_{e2}|^2 = 1$ and $U_{e1}U_{\mu 1}^* + U_{e2}U_{\mu 2}^* = 0$.)

- d) In a 2-flavor neutrino system, you may also write the mixing matrix as

$$U = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}. \quad (6)$$

In addition, for neutrinos we have $t \approx L$, where L is the distance traveled by the neutrino (recall $c = 1$). Write your result in terms of θ , the mass-squared differences Δm_{21}^2 , L and $E_\nu \simeq p$. Express your answer in terms of sin functions only.

- e) What happens if neutrinos were all massless? What about if $\Delta m_{21}^2 = 0$? Explain the physical intuition behind your result.
- f) Why happens for $\theta = 0, \pi/4, \pi/2$? Again explain what is happening in each case.



Question 2) Real data: Fig. 1 shows data taken by the Jiangmen Underground Neutrino Observatory (JUNO) in China and released very recently. It very clearly shows the effect of $\bar{\nu}_e$ disappearance due to neutrino oscillations (compare dashed black with the filled red histogram, or look at the ratio between the two at the bottom panel). Most of the disappearance can be attributed to a single mixing angle called θ_{12} and the corresponding mass-squared difference Δm_{21}^2 .

- a) Without looking this up, can you estimate the values of θ_{12} and Δm_{21}^2 from the data? Give your answer in degrees and eV^2 , respectively.
- b) Looking closely at the data, you can see a tiny-amplitude wiggles on the main red histogram. What could be causing this feature?

(This is a modern version of S&N Problem 2.4.)

Hint: You may find it useful: $\hbar c = 197 \text{ MeV fm} = 1$.

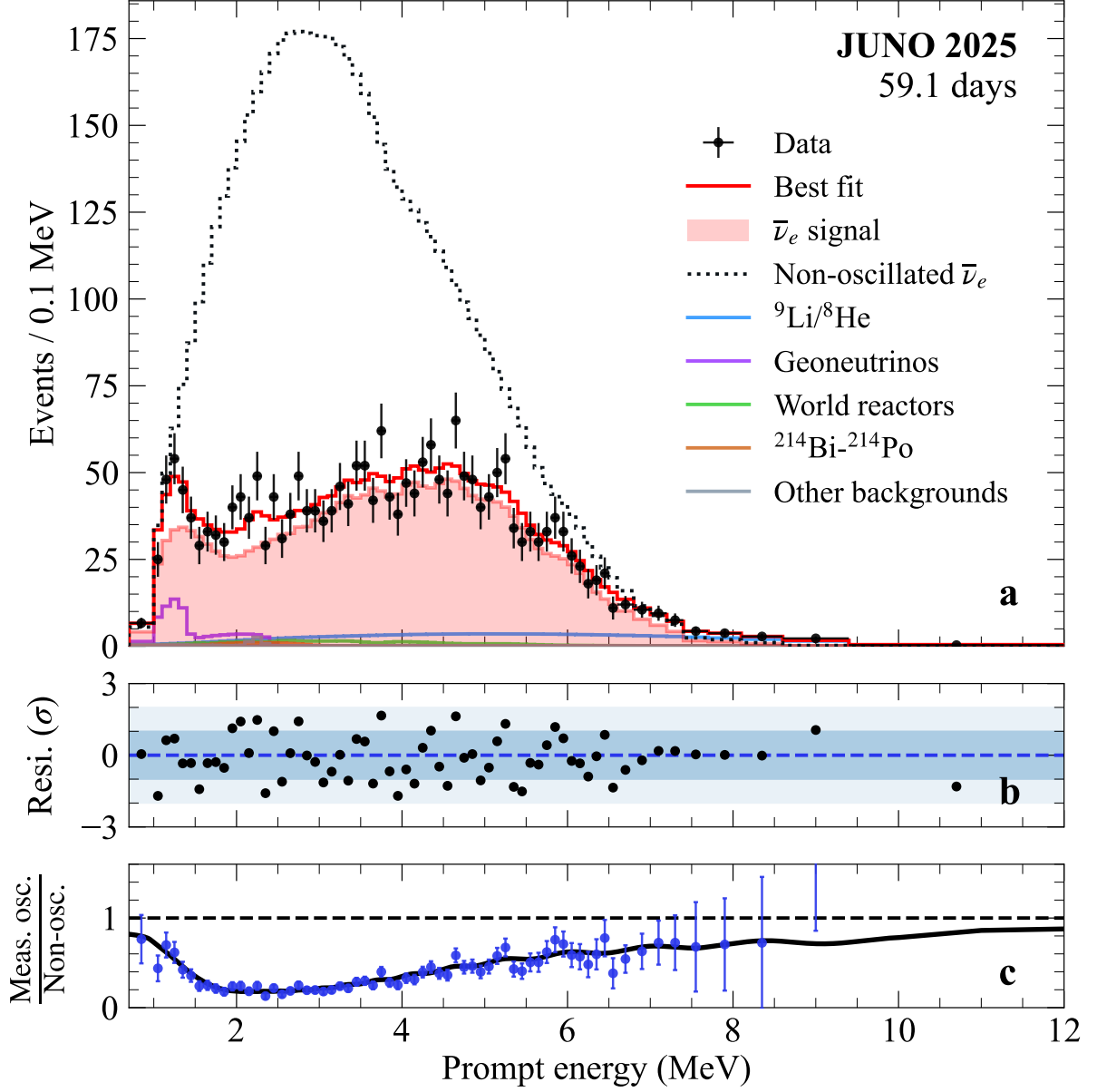


Figure 1: The energy spectrum of electron-antineutrinos ($\bar{\nu}_e$) at the JUNO detector in Jiangmen, China as measured with only 59 days of data. Each bin counts the number of events of a given energy. Data points are the number of observed events, and the filled red histogram represents the theory prediction of the electron-antineutrino energy spectrum. Antineutrinos are produced by fission reactions in ginormous nuclear power plants, all at an approximately equal distance from the detector of $L = 53.0$ km. Each event corresponds to an antineutrino that traveled from the reactors to the detector and underwent the reaction $\bar{\nu}_e + p \rightarrow e^+ + n$ with the e^+ being the detected particle. Each bin counts the number of neutrinos of a given energy in MeV (treat “prompt energy” $\approx E_\nu - 0.8$ MeV). See [2511.14593](#) for more details.



Bonus Question) (Remember **Mathematica** is your friend here.) The 3-neutrino flavor mixing matrix can be written in the following form:

$$\begin{aligned}
 U &= \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}
 \end{aligned} \tag{7}$$

with $c_{ij} = \cos \theta_{ij}$ and $s_{ij} = \sin \theta_{ij}$, and δ being a real number, also called the CP-violating phase.

- Calculate the three-neutrino oscillation probabilities for the two channels $P(\nu_\mu \rightarrow \nu_e)$ and $P(\nu_\mu \rightarrow \nu_\tau)$.
- Knowing that $\theta_{13} \simeq 0.1$ rad is small but $\theta_{23} \simeq \frac{\pi}{4}$ is large, which probability is largest for the Deep Underground Neutrino Experiment (DUNE): $L = 1300$ km and $E_\nu \sim 2$ GeV? Just for this comparison, you may set $\Delta m_{21}^2 \rightarrow 0$ (thereby reducing the problem to a two-neutrino system).
- DUNE's beam is almost purely made of ν_μ , so at the location of its detector, what is the primary flavor composition of the beam?
- How do your probabilities change for antineutrinos, knowing that all you need to do is replace U with its complex conjugate U^* ?
- What is

$$\Delta P = P(\nu_\mu \rightarrow \nu_e) - P(\bar{\nu}_\mu \rightarrow \bar{\nu}_e) \tag{8}$$

and under what conditions can it be nonzero? A non-zero observation of ΔP would prove that Charge-Parity (CP) symmetry is broken by leptons in the Standard Model.

1 Solutions Question 1

a) Taking the Hermitian conjugate of ?? gives $\langle \nu_\alpha | = \sum_i U_{\alpha i} \langle \nu_i |$, so

$$\begin{aligned} \langle \nu_\alpha | \nu_\beta \rangle &= \sum_{i,j} U_{\alpha i} U_{\beta j}^* \langle \nu_i | \nu_j \rangle = \sum_{i,j} U_{\alpha i} U_{\beta j}^* \delta_{ij} = \sum_i U_{\alpha i} U_{\beta i}^* \\ &= \sum_i U_{\alpha i} (U^\dagger)_{i\beta} = (UU^\dagger)_{\alpha\beta} = \delta_{\alpha\beta}, \end{aligned} \quad (9)$$

where I used $(U^\dagger)_{i\beta} = U_{\beta i}^*$. The flavor states form an orthonormal basis just as the mass states. Note that this direction used $UU^\dagger = \mathbb{1}$. Inverting ??, $|\nu_i\rangle = \sum_\alpha U_{\alpha i} |\nu_\alpha\rangle$, and checking that the mass states come out orthonormal would rely on $U^\dagger U = \mathbb{1}$ instead. For a square matrix the two conditions are equivalent.

b) Since the $|\nu_i\rangle$ are eigenstates of $\hat{H}$, the time-evolution operator $e^{-i\hat{H}t}$ acts on each of them as a complex phase:

$$|\nu_\alpha(t)\rangle = e^{-i\hat{H}t} |\nu_\alpha\rangle = \sum_i U_{\alpha i}^* e^{-iE_i t} |\nu_i\rangle. \quad (10)$$

c) The amplitude can quite generically be written as

$$A_{\alpha\beta}(t) \equiv \langle \nu_\beta | \nu_\alpha(t) \rangle = \sum_{i,j} U_{\beta j} U_{\alpha i}^* e^{-iE_i t} \langle \nu_j | \nu_i \rangle = \sum_i U_{\beta i} U_{\alpha i}^* e^{-iE_i t}, \quad (11)$$

so that for **two neutrinos**:

$$P_{\alpha\beta}(t) = |A_{\alpha\beta}(t)|^2 = |U_{\beta 1} U_{\alpha 1}^* e^{-iE_1 t} + U_{\beta 2} U_{\alpha 2}^* e^{-iE_2 t}|^2 = |U_{\beta 1} U_{\alpha 1}^* + U_{\beta 2} U_{\alpha 2}^* e^{-i(E_2 - E_1)t}|^2 \quad (12)$$

where I factored out the common phase $e^{-iE_1 t}$ in the last step, which gets squared to one. This is already technically enough to numerically evaluate the probability. Note how it only depends on the energy difference, which under our assumption of equal momenta for different mass eigenstates gives

$$\Delta E_{21} = E_2 - E_1 = \frac{m_2^2 - m_1^2}{2p} = \frac{\Delta m_{21}^2}{2p}. \quad (13)$$

The oscillation frequency is set by the energy difference.

We can use a neat trick to rewrite the above using unitarity of the mixing matrix: $U_{\alpha 1} U_{\beta 1}^* + U_{\alpha 2} U_{\beta 2}^* = \delta_{\alpha\beta}$, so that

$$P_{\alpha\beta}(t) = |\delta_{\alpha\beta} - U_{\beta 2} U_{\alpha 2}^* (1 - e^{-i\Delta E_{21} t})|^2, \quad (14)$$

so that for what we call an **appearance** probability (i.e., $\alpha \neq \beta$),

$$P_{\alpha\beta}(t) = |U_{\beta 2}|^2 |U_{\alpha 2}|^2 |1 - e^{-i\Delta E_{21} t}|^2 = 4 |U_{\beta 2}|^2 |U_{\alpha 2}|^2 \sin^2 \left(\frac{\Delta E_{21} t}{2} \right). \quad (15)$$

and for a **survival** probability (i.e., $\alpha = \beta$),

$$P_{\alpha\alpha}(t) = |1 - U_{\alpha 2} U_{\alpha 2}^* (1 - e^{-i\Delta E_{21} t})|^2 = 1 - 4 |U_{\alpha 1}|^2 |U_{\alpha 2}|^2 \sin^2 \left(\frac{\Delta E_{21} t}{2} \right). \quad (16)$$

d) The mixing matrix gives $|U_{e1}|^2 = \cos^2 \theta$ and $|U_{e2}|^2 = \sin^2 \theta$, so $4|U_{e1}|^2|U_{e2}|^2 = 4 \sin^2 \theta \cos^2 \theta = \sin^2 2\theta$. Setting $t \rightarrow L$ and $p \rightarrow E_\nu$,

$$\boxed{P_{\mu e}(L) = \sin^2(2\theta) \sin^2\left(\frac{\Delta m_{21}^2 L}{4E_\nu}\right), \quad P_{ee}(L) = 1 - \sin^2(2\theta) \sin^2\left(\frac{\Delta m_{21}^2 L}{4E_\nu}\right).} \quad (17)$$

These are the forms you would find on standard references for neutrino oscillations.

e) If all neutrinos were massless, no oscillations occur because $\Delta E_{21} = 0$, none of the neutrinos can develop a phase faster than the others. Relativistically, this is the notion that particles traveling at the speed of light do not experience any “proper” time, and therefore should not experience any dynamical effects like neutrino oscillations. The rest follows from a simple substitution into the desired formula.

f) Special attention to the case $\theta = 0$ or $\theta = \pi/2$, where the flavor states are identified with a single mass eigenstate. Since mass eigenstates are **stationary states** of the Hamiltonian, they do not change with time.